RAJALAKSHMI ENGINEERING COLLEGE

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B2

Started on	Tuesday, 15 September 2026, 8:42 AM
State	Finished
Completed on	Tuesday, 15 September 2026, 8:43 AM
Time taken	1 min 42 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array `nums` of size `n`, return *the majority element*.

The majority element is the element that appears more than `⌊n / 2⌋` times. You may assume that the majority element always exists in the array.

Example 1:

```
Input: nums = [3,2,3]
Output: 3
```

Example 2:

```
Input: nums = [2,2,1,1,1,2,2]
Output: 2
```

Constraints:

- `n == nums.length`
- `1 <= n <= 5 * 10^4`
- `-2^31 <= nums[i] <= 2^31 - 1`

For example:

Input	Result
3 3 2 3	3
7 2 2 1 1 1 2 2	2

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int majority(int a[], int l, int r) {
    if (l == r)
        return a[l];

    int mid = (l + r) / 2;
    int left = majority(a, l, mid);
    int right = majority(a, mid + 1, r);

    if (left == right)
        return left;

    int lc = 0, rc = 0;

    for (int i = l; i <= r; i++) {
        if (a[i] == left) lc++;
        if (a[i] == right) rc++;
    }

    return lc > rc ? left : right;
}

int main() {
    int n, a[50000];

    scanf("%d", &n);

    for (int i = 0; i < n; i++)
        scanf("%d", &a[i]);

    printf("%d", majority(a, 0, n - 1));

    return 0;
```

```
35 }
```

	Input	Expected	Got	
✔	3 3 2 3	3	3	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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